iBAS++

API DOCUMENTATION ON PWD-BIMH SYSTEM & PAIMS ITEGRATION

Scheme on Improvement of Public Financial Services Delivery through Implementation of BACS and iBAS++

Strengthening Public Financial Management Program to Enable Service Delivery (SPFMS)
Finance Division, Ministry of Finance

Document Classification: Restricted

Document Control

A. Document Information

Document Name	API Document for PWD-BIMH System
Document Number	iBAS/L3/
Version	2.0
Document Author	Chief Information Security Officer (CISO)
Approval Date	
Release Date	
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Review Frequency	This document and its content statements will be reviewed on an as-needed basis.
Document Reference	ISO/IEC 27001:2013 Clause- A.14

B. Document Distribution List

Serial No	Person / Department / Designation / Team
01	Security Team
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C. Author Information

Prepared By	**Name:** Shakila Rahman **Designation**: IT Consultant (System Documentation) BACS and iBAS++ Scheme, SPFMS program, Finance Division, Ministry of Finance Date:	
Reviewed By	Name: Ash-Shakur Rahaman Designation: Sr. IT Consultant BACS and iBAS++ Scheme, SPFMS program, Finance Division, Ministry of Finance Date:	
Recommended By	Name: Abu Saleh Zainal Abedin Designation: Team Lead (IT) BACS and iBAS++ Scheme, SPFMS program, Finance Division, Ministry of Finance Date:	
Approved By	**Name**: Muhammad Anisuzzaman **Designation**: Program Executive & Coordinator (Deputy Secretary) BACS and iBAS++ Scheme, SPFMS program, Finance Division, Ministry of Finance **Date**:	

D. Document History

Version No	Revision Date	Approval Date	A/M/ D/N	Description of Change	Document Owner	Approved By
1.0	N/A		N	First Release	iBAS++ Scheme	National Program Director

*A= Added, M= Modified, D= Deleted, N

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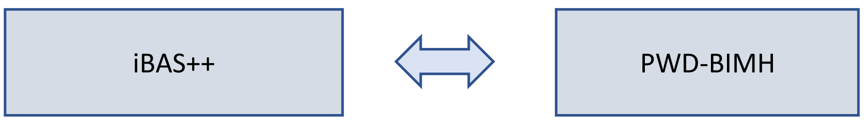
1 INTRODUCTION

The Public Works Department (PWD) plays a major role in executing government structural development projects throughout the financial year. At the start of each financial year, PWD formulates its development budget in alignment with the government's structural development plan. Based on the estimated budget, projects are rolled out. Establishing integration with the iBAS++ system, effectively connecting with the PWD-BIMH system, is vital for real-time updates and monitoring. This facilitates the implementation of a comprehensive and automated budget management system.

This integration will ensure various advantages, contributing to improved tracking of the update and increased financial efficiency, as detailed below.

- Streamlined Execution: PWD's major role in executing government structural development projects is streamlined throughout the financial year.
- Aligned Development Plan: PWD's development plan is formulated at the beginning of each financial year, aligning with the government's structural development budget.
- Effective Project Rollout: Projects are rolled out based on the estimated budget, ensuring effective and efficient execution.
- Real-time Updates and Monitoring: Integration with the iBAS++ system allows for real-time updates and monitoring of projects, enhancing overall project management.
- Comprehensive Budget Management: The integration facilitates the implementation of a comprehensive and automated budget management system, optimizing financial processes and resource allocation.



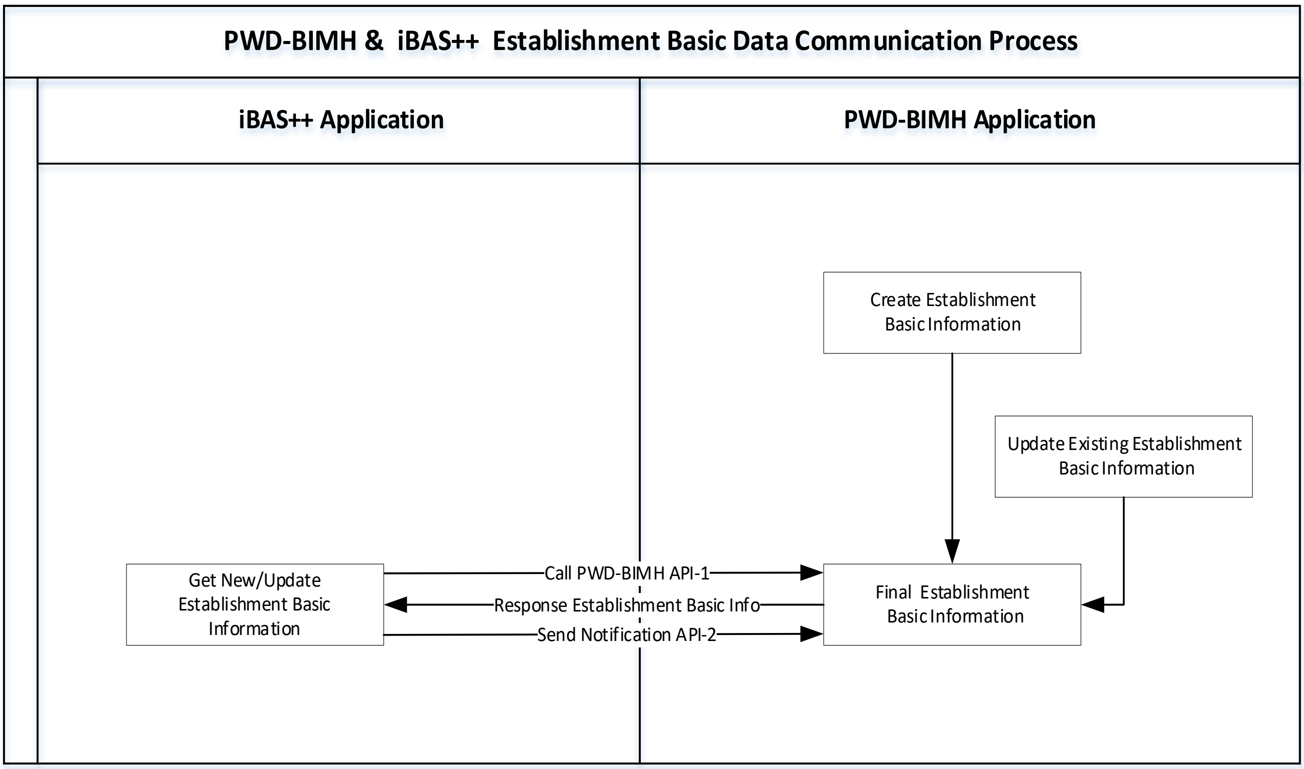
2 AUDIENCE

This document primarily targets the IT development teams associated with both systems—those interfacing with the PWD-BIMH and iBAS++ system's development. It serves as a comprehensive guide for this development API DETAILS & STEPS.

3 API Sharing Process Steps

- Initialization: To ensure that all essential components and parameters in the API request.
- Authentication: Implement a secure authentication mechanism to validate the source and destination to enhance data security.
- Data Preparation: Organize and prepare the data to be pushed, ensuring it complies with the specified format and requirements.
- Parameter Inclusion: Incorporate essential parameters such as fiscal year ID, token number, and system control point ID to contextualize and identify the data being shared.
- Error Handling: Will be Implementation of robust error-handling mechanisms to manage and communicate any issues that may arise during the API sharing process.
- Verification: Will be Conduct post-push verification to ensure the accurate and complete transfer of data between systems.
- Required input parameter will need to be provided during API call and response will be in JSON format.

4 Workflow Diagram



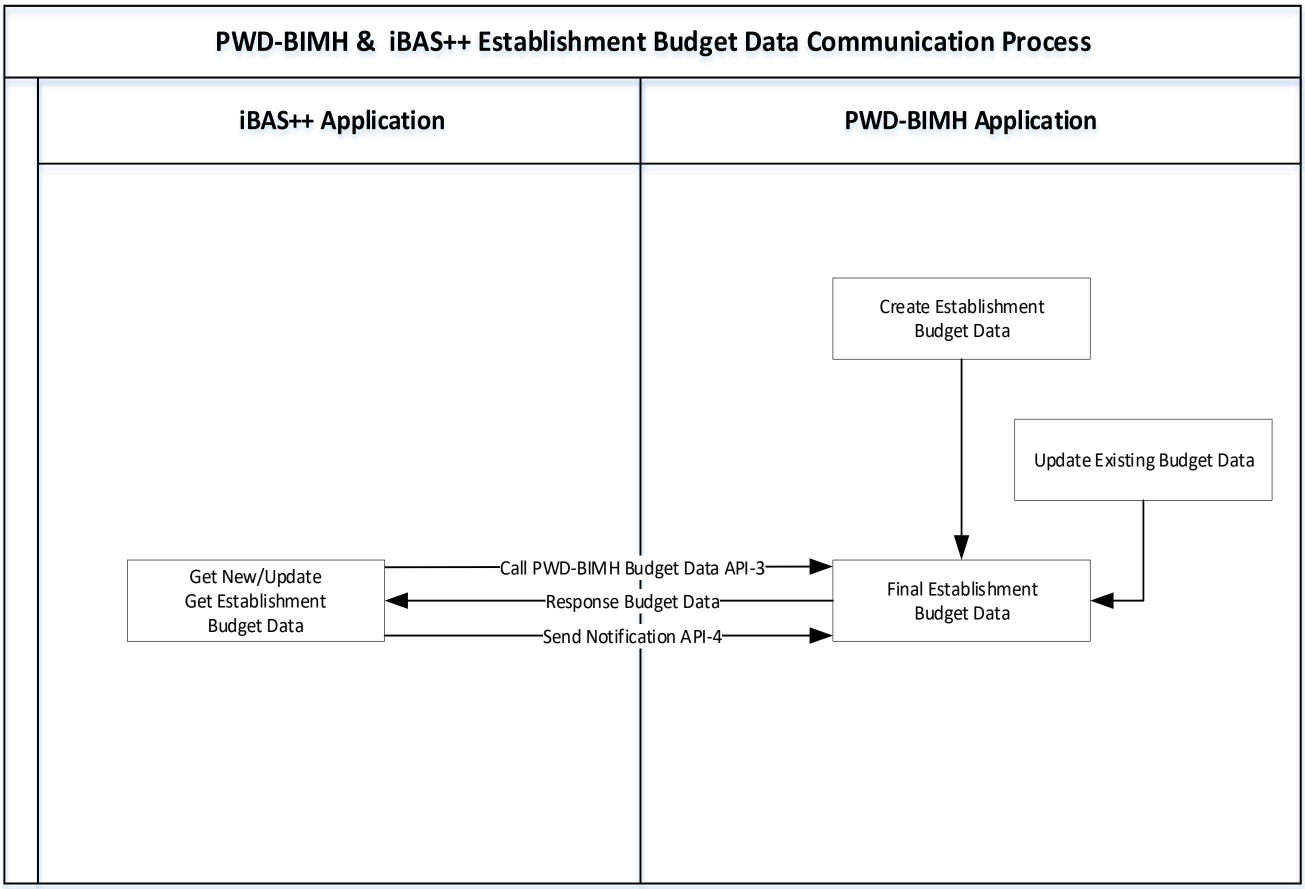


Figure: PWD-BIMH & iBAS++ System Data communication Cross-functional Diagram

5 API DETAILS & STEPS

Response Mapping Table: To identify by a numeric code (Response Code) of the response call following is the mapping table. Both Programme MIS & Single Registry System will need to use this table.

PKID	Response Code	Response Description
1	0	Successfully Processed
2	1	Invalid Input Parameter Request
3	2	Token Generation failed
4	3	User authentication failed. Wrong User ID or Password
5	4	User authentication failed. Old Password Mismatch
6	5	Token Expired
7	6	Transaction ID mismatch or not found

[Please suggest from your end to incorporate the response. We shall include this accordingly.]

5.1 Authorization/Token

Considering security when an API call will be made from iBAS++ or PWD-BIMH, a unique access token on that call will be supplied for that username and password. This token acts like an electronic key to access to the API. This access token is required for each API call.

5.1.1 Summary

API Name	**Authorization**
Details	Authenticate user for token generation
Method	POST
Access to	PWD-BIMH System
Request object	Restful Web API
Response	Json formatted data
Method Name/URL	{base-Url}/api/token
Header Parameter	Authorization:Basic VENNOkVFRjQ3RDlBLURCQTktNEQwMi1CN0IwLTA0RjQyNzlBNkQyMA== Content-Type:application/x-www-form-urlencoded

Request format	{
	grant_type : '******',
	username : ‘******’,
	Password : ‘******’
	}

Response format	Json data

Sample Request	{
	grant_type : '******',
	username : '******',
	Password : '******'
	}

Sample Response	{ "access_token" : "rP- vkJYGa0qIseiF8F6vKWFKtqycPvBSlwr2iaAxYrMqKK1wieZcPplfS566muD4ofmH9BYgSTRpLxUJQTAT_PmRQ MsyxRajaJKmTezPs2HskJVKGbzGcAbq2t1aurfgDWsHbjN3MnWTAz0Gf97SgGJOb2zjgoI6P4fbOa5Q4RqaEG ok9dC0wv5g2FWkTErcfK0TAUUYtjCsWUV_BcyqgR730GMnnR5FzBumCGP9QUOItooBO9moFlu9wVr6hMx WF1xOs8i48CnfmzoUuAcpUraMV2wuqkEYcjlONZgJL8cRTxJEaGg2kJXWDSMWyEv8", "token_type" : "bearer", "expires_in" : 599, "refresh_token" : "********", "client_id" : "******", "userName" : "******", "issued" : "Thu, 06 Feb 2020 04:00:06 GMT", "expires" : "Thu, 06 Feb 2020 04:10:06 GMT",

API Name	**Authorization**
	"responsecode" : "0" }

5.1.2 Request Parameters

#	Name	Data Type	Constrains	Detailed Description
1	Body: grant_type	String	Required	Authentication type (i.e. refresh token/password)
2	Body: username	String	Required	User name for getting token
3	Body: password	String	Required	Password of the user name.
4	Header: authorization	String	Required	Basic + hash value key
5	Header: content-type	String	Required	application/x-www-form-urlencoded

5.1.3 Response Parameters

#	Name	Data Type	Detailed Description
1	Access Token	String	If request is valid, the authorization server generates an access token and return these to the client which allows client to access API without re-entering the user's credentials.
2	Token Type	String	It represents the type of the token that need to add in the header.
3	Expires In	String	It indicates the lifetime in seconds of the access token. For example, the value "3600" denotes that the access token will expire in one hour from the time the response was generated.
4	Refresh Token	String	It is used to obtain a new access token when the current access token becomes invalid or expires.
5	ClientID	String	The ClientID is a public identifier for apps.
6	Username	String	User Name provided to access this API
7	Issued	Date	It indicates the issue date &time of that access token
8	Expires	date	It indicates the expire date & time of that access token
9	Response code	String	Response Code will be returned. Like: 0 - Successfully Processed 1- Invalid Input Parameter Request 2 - Token Generation failed 3 - User authentication failed. Wrong User ID or Password

5.2 Change API Password

Through this API password can be changed. To do this old and new password will need to be provided to the API.

5.2.1 Summary

API Name	**Change API Password**
Details	Access for change password information
Method	GET
Access to	PWD-BIMH System
Request object	Restful Web API
Response	Json formatted data
Method Name/URL	{base-URL}/api/IbasAPIService/ChangeApiPassword

Header Parameter	Content-Type: application/json Authorization:Bearer rP-vkJYGa0qIseiF8F6vKWFKtqycPvBSlwr2iaAxYrMqKK1wieZcPplfS566muD4ofmH9BYgSTRpLxUJQTAT_PmRQMsyxRajaJKmTezPs2HskJVKGbzGcAbq2t1aurfgDWsHbjN3MnWTAz0Gf97SgGJOb2zjgoI6P4fbOa5Q4RqaEGok9dC0wv5g2FWkTErcfK0TAUUYtjCsWUV_BcyqgR730GMnnR5FzBumCGP9QUOltooBO9moFlu9wVr6hMxWF1xOs8i48CnfmzoUuAcpUraMV2wuqkEYcjlONZgJL8cRTxJEaGg2kJXWDSMWyEv8
Request format	'Authorization': 'Bearer ' + access_token
Response format	Json data

Sample Request	{
	"userName" : "******",
	"oldPassword" : "******",
	"newPassword" : "******"
	}

Sample Response	{
	"responsecode" : "0"
	}

5.2.2 Request Parameters

#	Request Parameter Name	Data Type	Constrains	Detailed Description
1	Param: user Name	String	Required	User name
2	Param: old Password	String	Required	Old Password of the user name.
3	Param: new Password	String	Required	New Password of the user name
4	Header Parameter: Authorization	String	Required	Bearer + white space + access token

5.2.3 Response Parameters

#	Response Name	Data Type	Detailed Description
1	Response code	String	Response Code will be returned. Like: 0-Successfully Processed 11-Invalid Input Parameter Request

5.3 PWD-BIMH Establishment Basic Data API

The iBAS++ system will send a request to PWD-BIMH to receive Establishment basic data. Further details regarding this request are discussed below.

5.3.1 Summary

API Name	**PWD-BIMH Establishment Basic Data API**
Details	This API will be used to send requests to PWD-BIMH to receive Establishment basic data.
Method	Get
Access to	PWD_BIMH System to iBAS++ System
Request object	Restful Web API
Response	JSON formatted data
URL	{base-url}/api/PWD_BIMH/GetPWDEstablishmentBasicData
Header Parameter	

Request format	{
	requestId String(32),
	ministryCode String (3)
	}

Response format	{ requestId String(32), [{

<table>
<tr><td></td><td>establishmentId Number(18),
establishmentNameEn String(1000),
establishmentNameBn String(2000),
ministryCode String (3),
pwddivisionCode String (13),
projectNameEn String (1000),
projectNameBn String (2000),
establishmentType String(50),
locationCode String(9),
yearofConstruction Number(5),
approximately Number(5),
noofFloorAboveGround Number(5),
noOfFloorUnderGround Number(5),
noOfFloorParking Number(5),
establishmentHeightUoM String(20),
establishmentHeight Number(5),
plinthAreaUoM String(20),
plinthArea Number(5),
totalFloorAreaUoM String(20),
totalFloorArea Number(5),
acNo Number(5),
totalACCapacityUoM String (20),
totalACCapacity Number(5),
generatorNo Number(5),
totalGeneratorCapacityUoM String (20),
totalGeneratorCapacity Number(5),
substationNo Number(5),
totalSubstationCapacityUoM String (20),
totalSubstationCapacity Number(5),
fireDetectionSystem String(1000),
fireProtectionSystem String(1000),
liftNo Number(5),
emOtherInformation String(1000),
dataType String(1)
}
]
}</td></tr>
<tr><td>Sample Request</td><td>{
"requestId" : "************************",
"ministryCode" : "144"
}</td></tr>
<tr><td>Sample Response</td><td>{
“requestId” : "********************",
[
{
“establishmentId” : “0056456540”,
“establishmentNameEn“ : “Test Building”,
“establishmentNameBn“ : “টেস্ট বিল্ডিং”,
“ministryCode“ : “144”,
“pwddivisionCode“ : “4654654”,
“projectNameEn“ : “Test Project Building”,
“projectNameBn“ : “টেস্ট প্রজেক্ট বিল্ডিং”,
“establishmentType“ : “General Building”,
“locationCode“ : “152”,</td></tr>
</table>

	"yearofConstruction" : "2024", "approximately" : "2024", "noofFloorAboveGround" : "5", "noOfFloorUnderGround" : "1", "noOfFloorParking" : "2", "establishmentHeightUoM" : "ft", "establishmentHeight" : "5" "plinthAreaUoM" : "sq.ft", "plinthArea" : "65", "totalFloorAreaUoM" : "sq.ft", "totalFloorArea" : "20", "acNo" : "10001", "totalACCapacityUoM" : "TON", "totalACCapacity" : "10", "generatorNo" : "2", "totalGeneratorCapacityUoM" : "KVA", "totalGeneratorCapacity" : "2", "substationNo" : "1", "totalSubstationCapacityUoM" : "KVA", "totalSubstationCapacity" : "10", "fireDetectionSystem" : "Test Project Information", "fireProtectionSystem" : "Test1 Project Information", "liftNo" : "2", "emOtherInformation" : "Test1 Project Information", "dataType" : "N" }] }

Error	{
	"errorCode" : "500",
	"Badrequest" : "400"
	}

5.3.2 Request Parameters

#	Request	Data Type	Constrains	Detailed Description
1	Request Id	String	Required	32 Digit Characters
2	Ministry Code	Numbers	Required	3 Digit Ministry Code

5.3.3 Response Parameters

#	Response	Data Type	Constrains	Detailed Description
1	Request Id	String	Required	32 Digit Characters
2	Establishment ID	Number	Required	18 Digit Numbers
3	Establishment Name(English)	String	Required	1000 Digit Characters
4	Establishment Name(Bangla)	String		2000 Digit Characters
5	Ministry Code	String	Required	3 Digit Characters
6	PWD Division Code	String	Required	13 Digit Characters

#	Response	Data Type	Constrains	Detailed Description
7	Project Name (English)	String	Required	1000 Digit Characters
8	Project Name (Bangla)	String		2000 Digit Characters
9	Establishment Type	String	Required	20 Digit Characters, Establishment Type will be Residential and Non-residential
10	Location Code	String	Required	9 Digit Characters, It will be the shared location of the establishment.
11	Year of Construction	Number	Required	5 Digit Numbers
12	Approximately Number	Number		5 Digit Numbers
12	No. of Floor Above Ground	Number		5 Digit Numbers
14	No. of Floor Under Ground	Number		5 Digit Numbers
15	No. of Floor Parking	Number		5 Digit Numbers
16	Establishment Height UoM	String		20 Digit Characters
17	Establishment Height	Number		5 Digit Numbers, Establishment Height specified in feet format
18	Plinth Area UoM	String		20 Digit Characters
19	Plinth Area	Number		5 Digit Numbers, Plinth area is specified in square feet format
20	Total Floor Area UoM	String		20 Digit Characters
21	Total Floor Area	Number	Required	5 Digit Numbers, Total Floor area is specified in square feet format
22	AC No	Number	Required	5 Digit Numbers
23	Total AC Capacity UoM	String		20 Digit Characters
24	Total AC Capacity	Number		5 Digit Numbers
25	Generator No	Number		5 Digit Numbers
26	Total Generator Capacity UoM	String		20 Digit Characters
27	Total Generator Capacity	Number		5 Digit Numbers
28	Total Substation No	Number		5 Digit Numbers
29	Total Substation Capacity UoM	String		20 Digit Characters
30	Total Substation Capacity	Number		5 Digit Numbers
31	Fire Detection System	Number		5 Digit Numbers
32	Fire Protection System	Number		5 Digit Numbers
33	Lift No	Number		5 Digit Numbers
34	EM Other Information	String		1000 Digit Characters
35	Data Type	String	Required	1 Digit Character, provided Establishment Data is new, Data Type is 'N';if the Data is modified or updated then the Data Type is 'U'.

5.4 PWD-BIMH Basic Data insert Notification API

Redirected to the PWD-BIMH system, upon successfully receiving basic Establishment data. Further details regarding this request are discussed below.

5.4.1 Summary

API Name	**PWD-BIMH Basic Data insert Notification API**
Details	Redirected to the PWD-BIMH system after successfully receiving basic Establishment data.
Method	Post
Access to	iBAS++ System to PWD-BIMH System
Request object	Restful Web API
Response	
URL	/PWDBIMH/IbasInsertComplete
Response	
Query Parameter	
Response	
Request Body	{ “establishmentId” : [“5645654”,”64765456”,”5454545”] }
Sample Response	} “Successful” : “true” }
Error	{ "errorCode" : "500", "Badrequest" : "400" }

5.4.2 Request Parameters

#	Request	Data Type	Constrains	Detailed Description
1	Establishment ID	Number	Required	18 Digit Numbers

5.4.3 Response Parameters

#	Name	Data Type	Constrains	Detailed Description
1	Successful	Boolean	Required	If the user request is successful, the response will be "true"; if the user is not successful, the response will be "false".

5.5 PWD-BIMH Establishment Budget Data API

iBAS++ system will send a request to PWD-BIMH to receive Establishment Budget data. Further details regarding this request are discussed below.

5.5.1 Summary

API Name	**PWD-BIMH Establishment Budget Data API**
Details	This API will be used to send requests to PWD-BIMH to receive Establishment Budget data.
Method	Get
Access to	PWD_BIMH System to iBAS++ System
Request object	Restful Web API

Response	JSON formatted data
URL	{base-url}/api/PWD_BIMH/GetPWDBudgetData
Header Parameter	

Request format	{
	requestId String (32),
	ministryCode String (3),
	fiscalYear String (18),
	budgetPhase String(20)
	}

Response format	{
	requestId String (32),
	ministryCode String (3),
	fiscalYear String (18)
	[
	{
	budgetPhase String(20)
	establishmentId Number(18),
	establishmentType String (20),
	amount Number(18) Default 0,
	amountProj1 Number(18) Default 0,
	amountproj2 Number(18) Default 0
	}
	]
	}

Sample Request	{
	“requestId” : “******************”,
	“ministryCode” : “144”,
	“fiscalYear” : “2024-25”,
	“budgetPhase” : “Budget”
	
	}

Sample Response	{
	“requestId” : "**********************",
	“ministryCode” : “144”,
	“fiscalYear” : “2024-25“
	“budgetPhase” : “Budget”,
	
	[
	{
	“establishmentId” : “0056456540”,
	“establishmentType” : “Economic Code of 7 digit”,
	“amount” : “10000000”,
	“amountProj1” : “10000000”,
	“amountproj2” : “10000000”
	}
	]
	}

Error	{
	"errorCode" : "500",
	"Badrequest" : "400"
	}

5.5.2 Request Parameters

#	Request	Data Type	Constrains	Detailed Description
1	Request Id	String	Required	32 Digit Characters
2	Fiscal year	String	Required	32 Digit String
3	Ministry Code	String	Required	3 Digit Characters
4	Budget Phase	String	Required	32 Digit Characters, Budget Phase will be 'Budget' or 'Revised'

5.5.3 Response Parameters

#	Response	Data Type	Constrains	Detailed Description
1	Request Id	String	Required	32 Digit Characters
2	Fiscal year	String	Required	32 Digit String
3	Ministry Code	String	Required	3 Digit Characters
4	Establishment ID	Number	Required	18 Digit Numbers
5	Budget Phase	String	Required	32 Digit Characters, Budget Phase will be 'Budget' or 'Revised'
6	Establishment Type	String	Required	7 Digit Characters, Establishment Type will be Residential and Non-residential which will be presented as Economic Code of 7 digit.
7	Amount	Number	Required	18 Digit Numbers, Default 0. It will be the budgetary amount.
8	Amount proj1	Number	Required	18 Digit Numbers, Default 0. It will be the amount projected for the first year
9	Amount proj2	Number		18 Digit Numbers, Default 0. It will be the amount projected for the second year.

5.6 PWD-BIMH Establishment Budget Data insert Notification API

Redirected to the PWD-BIMH system, upon successfully receiving Establishment Budget data. Further details regarding this request are discussed below.

5.6.1 Summary

API Name	**PWD-BIMH Basic Data insert Notification API**
Details	Redirected to the PWD-BIMH system after successfully receiving Establishment Budget data.
Method	Post
Access to	iBAS++ System to PWD-BIMH System
Request object	Restful Web API
Response	
URL	/PWDBIMH/IbasEstablishmentBudgetInsert
Response	

Request Body	{
	 establishmentId Number(18),
	 }

Response	

Sample Request	{ "establishmentId" : ["5645654","64765456","5454545"] }
Sample Response	} "successful" : "true" }
Error	{ "errorCode" : "500", "Badrequest" : "400" }

5.6.2 Request Parameters

#	Name	Data Type	Constrains	Detailed Description
1	Establishment ID	Number	Required	18 Digit Numbers

5.6.3 Response Parameters

#	Name	Data Type	Constrains	Detailed Description
1	Successful	Boolean	Required	If the user request is successful, the response will be "true"; if the user is not successful, the response will be "false".

6 API for PAIMS

6.1 PWD-BIMH Establishment Asset Data API

The Public Asset Information Management System (PAIMS) will send a request with a unique ID to PWD-BIMH to receive Establishment basic data for Public Asset Register. For the 1st time, PAIMS will get all Establishment approved Data from PWD-BIMH system, where PWD-BIMH will mark those data with delta mark to update the status as those are sent to PAIMS system.
To get the updated data, PAIMS will run scheduler on daily/ weekly/ monthly basis as per management requirement. At that point PWD-BIMH will send the latest data accordingly.
Further details regarding this request are discussed below.

6.1.1 Summary

API Name	**PWD-BIMH Establishment Asset Data API**
Details	This API will be used to send requests to PWD-BIMH to receive Establishment profile data for Asset registry system.
Method	Get
Access to	PWD_BIMH System to PAIMS System
Request object	Restful Web API
Response	JSON formatted data
URL	{base-url}/api/PWD_BIMH/GetPWDEstablishmentAssetData
Header Parameter	
Request format	{ requestId String(32) }
Response format	{ requestId String(32), [{ establishmentId Number(18),

	establishmentNameEn String(1000), establishmentNameBn String(2000), projectNameEn String(1000), projectNameBn String(2000), concernedMinistry String(200), constructedBy String(200), administrativeDivision String(100), administrativeDistrict String(100), administrativeUpazila String(100), administrativeUnion String(100), latitude String(50), longitude String(50), buildingCategory String(1), // A, B, C, or D constructionCompletionDate String(10), // Format: YYYY-MM-DD usesOfEstablishment String(100), establishmentHeight Number(5), // Height value establishmentHeightUoM String(20), // Unit: meter/feet aboveGroundStoreys Number(3), hasBasement Boolean, basementLevels Number(2), plinthArea Number(7), plinthAreaUoM String(20), builtUpArea Number(7), builtUpAreaUoM String(20), structureType String(100), // RC Frame, Steel, etc. foundationType String(100), // Mat, Pile, Footing foundationDesignStorey Number(3), liftNumber Number(3), acNumber Number(3), acCapacity Number(5), acCapacityUoM String(20), // Tons motorNumber Number(3), motorCapacity String(20), // e.g., "10 HP" substationNumber Number(3), substationCapacity String(20), // e.g., "500 KVA" generatorNumber Number(3), generatorCapacity String(20), // e.g., "300 KVA" fireDetectionSystem String(1000), fireProtectionSystem String(1000), establishmentUser String(200), costOfConstruction Number(12), // in BDT sourceOfFund String(100), // ADP, GoB, Donor-funded landArea Number(10), landAreaUoM String(20), // decimals/acre mouza String(200), khatiyan String(200), landCertificate Boolean, remainingLife Number(3), // in years valuation Number(12), maintenanceAuthority String(200), buildingCondition String(10), // Good, Fair, Poor maintenanceType String(50), // Regular, Special, Day-to-Day maintenanceRequired Boolean, estimatedCostForMaintenance Number(12), remarks String(1000),

	establishmentType String(50), // Residential, Non-residential useType String(100), // Official, Commercial, Hospital, etc. structureClass String(100), // RC Frame, Masonry, Semi-Pucca foundationClass String(100), // Pile, Mat, Footing ancillaryFacilities Array[String(100)] // List: Prayer Hall, Pump House, etc. }] }

Sample Request	{
	"requestId" : "*************************"
	}

Sample Response	{
	"requestId": "abcd1234efgh5678ijkl9012mnop3456",
	[
	{
	"establishmentId": "EST-00001",
	"establishmentName": "PWD Head Office",
	"projectName": "PWD Modernization Project",
	"concernedMinistry": "Ministry of Housing and Public Works",
	"constructedBy": "PWD",
	"administrativeLocation": {
	"division": "Dhaka",
	"district": "Dhaka",
	"upazila": "Ramna",
	"union": null
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	"gpsCoordinates": {
	"latitude": "23.7382",
	"longitude": "90.3952"
	},
	"buildingCategory": "A",
	"constructionCompletionDate": "2020-12-31",
	"usesOfEstablishment": "Official",
	"establishmentHeight": "45.5m",
	"aboveGroundStoreys": 10,
	"undergroundBasement": {
	"hasBasement": true,
	"levels": 2
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	"plinthArea": "2000 sqm",
	"builtUpArea": "15000 sqm",
	"structureType": "RC Frame",
	"foundationType": "Pile",
	"foundationDesignStorey": 12,
	"liftNumber": 4,
	"acNumber": 20,
	"acCapacity": "60 tons",
	"motorDetails": {
	"number": 3,
	"capacity": "10 HP"

	},
	"substationDetails": {
	"number": 1,
	"capacity": "500 KVA"
	},
	"generatorDetails": {
	"number": 2,
	"capacity": "300 KVA"
	},
	"fireDetectionSystem": "Smoke & Heat Detectors",
	"fireProtectionSystem": "Extinguishers, Sprinklers, Hydrants",
	"establishmentUser": "Ministry of Housing and Public Works",
	"costOfConstruction": 150000000,
	"sourceOfFund": "GoB",
	"landArea": "200 decimals",
	"mouza": "Ramna",
	"khatiyan": "12345",
	"landCertificate": true,
	"remainingLife": 35,
	"valuation": 180000000,
	"maintenanceAuthority": "PWD",
	"buildingCondition": "Good",
	"maintenanceType": "Regular",
	"maintenanceRequired": false,
	"estimatedCostForMaintenance": 0,
	"remarks": "No major issues.",
	"establishmentType": "Non-residential",
	"useType": "Official",
	"structureClass": "RC Frame",
	"foundationClass": "Pile",
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	"Prayer Hall",
	"Drainage",
	"Garage"
	]
	}
	]
	}

Error	{
	"errorCode" : "500",
	"Badrequest" : "400"
	}

6.1.2 Request Parameters

#	Request	Data Type	Constrains	Detailed Description
1	Request Id	String	Required	32 Digit Characters

6.1.3 Response Parameters

#	Response	Data Type	Constrains	Detailed Description
1	Establishment ID	Number	Required	Unique ID assigned to each building or structure.
2	Establishment Name	String	Required	Official name of the building or structure.
3	Project Name	String	Optional	Associated development project name.
4	Concerned Ministry	String	Required	Ministry responsible for the building.
5	Constructed By	String	Required	Executing agency (PWD, LGED, etc.).
6	Administrative Location	Object	Required	Includes division, district, upazila, and union.
7	└ Division	String	Required	Administrative Division name.
8	└ District	String	Required	Administrative District name.
9	└ Upazila	String	Required	Administrative Upazila name.
10	└ Union	String/null	Optional	Administrative Union name.
11	GPS Coordinates	Object	Required	Latitude and Longitude of the building.
12	└ Latitude	String	Required, decimal format	Latitude value.
13	└ Longitude	String	Required, decimal format	Longitude value.
14	Building Category	String	Required	Classification based on GoB standards.
15	Construction Completion Date	String	Required, YYYY-MM-DD	Date when construction was completed.
16	Uses of Establishment	String	Required	Functional usage (Residential, Official, etc.).
17	Establishment Height	String	Optional, with unit (m/ft)	Height of the structure.
18	Above Ground Storeys	Number	Optional	Number of floors above ground.
19	Underground/Basement	Object	Optional	Basement existence and level count.
20	└ hasBasement	Boolean	Optional	Indicates if basement exists.
21	└ levels	Number	Optional	Number of underground levels.
22	Plinth Area	String	Required	Area of building footprint.
23	Built-up Area	String	Optional, with unit	Total constructed floor area.
24	Structure Type	String	Optional	Type of structural frame (e.g., RC Frame).
25	Foundation Type	String	Optional	Foundation type (e.g., Pile, Mat).

#	Response	Data Type	Constrains	Detailed Description
26	Foundation Design Storey	Number	Optional	Storey capacity designed for the foundation.
27	Lift Number	Number	Optional	Total number of lifts installed.
28	AC Number	Number	Optional	Total number of air conditioners.
29	AC Capacity	String	Optional	Total cooling capacity.
30	Motor Number & Capacity	Object	Optional	Pump/motor count and individual capacity.
31	└ number	Number	Optional	Number of water motors.
32	└ capacity	String	Optional	Motor capacity per unit.
33	Substation Number & Capacity	Object	Optional	Electrical substation count and capacity.
34	└ number	Number	Optional	Number of substations.
35	└ capacity	String	Optional	Capacity per substation.
36	Generator Number & Capacity	Object	Optional	Backup generator details.
37	└ number	Number	Optional	Number of generators.
38	└ capacity	String	Optional	Capacity per generator.
39	Fire Detection System	String	Optional	Type and coverage of detection devices.
40	Fire Protection System	String	Optional	Extinguishers, sprinklers, hydrants, etc.
41	Establishment User	String	Optional	Ministry/agency using the building.
42	Cost of Construction	Number	Optional	Actual construction cost.
43	Source of Fund	String	Optional	Funding source (GoB, ADP, Donor).
44	Land Area	String	Optional	Land size (e.g., 120 decimals, 1 acre).
45	Mouza	String	Optional	Mouza name or identifier.
46	Khatiyan	String	Optional	Legal land registration number.
47	Land Certificate	Boolean	Optional	Whether land has a legal certificate.
48	Remaining Life	Number	Optional	Estimated remaining life.
49	Valuation	Number	Optional	Most recent valuation of the property.
50	Maintenance Authority	String	Optional	Responsible agency for upkeep.
51	Building Condition	String	Optional	Current structural condition.
52	Maintenance Type	String	Optional	Type of maintenance (Regular, Special, etc.).
53	Maintenance Required	Boolean	Optional	Whether maintenance is currently required.

#	Response	Data Type	Constrains	Detailed Description
54	Estimated Cost for Maintenance	Number	Optional	Estimated repair cost.
55	Remarks	String	Optional	Additional notes if any.
56	Establishment Type	String	Required	Residential or Non-residential.
57	Use Type	String	Optional	Official, Commercial, Storage, etc.
58	Structure Class	String	Optional	Repeated structural classification (e.g., RC Frame).
59	Foundation Class	String	Optional	Repeated foundation type (e.g., Pile, Mat).
60	Ancillary Facilities	Array of Strings	Optional	Extra facilities (e.g., Prayer Hall, Pump House, etc.).

7 CREDENTIALS

Base URL	****************
User Name	****************
Password	****************